

DRIVER ASSISTANCE PERCEPTION SYSTEM

High-Level Design Document

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Classification: Synthetic Demonstration Data

1. System Overview

The Driver Assistance Perception System coordinates its functions through typed AUTOSAR interfaces and ports. Software components exchange signals through provider and consumer ports.

2. Software Architecture

The primary software components are PerceptionManager, RadarControl, CameraControl, BrakeAssistControl, LaneKeepControl, and WarningManager.

3. Software Components

Component	Role	Primary Responsibility
PerceptionManager	Coordinator	Fuses object lists and publishes the environment model.
RadarControl	Provider	Delivers radar object distance and velocity.
CameraControl	Provider	Delivers lane and object detections.
BrakeAssistControl	Consumer	Requests braking when a collision is predicted.
LaneKeepControl	Consumer	Computes steering corrections.
WarningManager	Consumer	Raises driver warnings.

4. Interfaces

Interface	Provider	Consumer	Signals
IRadarObjects	RadarControl	PerceptionManager	ObjectDistance
ILaneDetection	CameraControl	PerceptionManager	LaneOffset
ICollisionRisk	PerceptionManager	BrakeAssistControl	CollisionRisk
ISteeringCorrection	LaneKeepControl	PerceptionManager	SteeringAngleCommand
IDriverWarning	PerceptionManager	WarningManager	WarningLevel
ICameraStatus	CameraControl	WarningManager	CameraStatus

5. Ports

Component	Port	Direction	Interface
RadarControl	RadarObjects_Out	P-PORT	IRadarObjects
PerceptionManager	RadarObjects_In	P-PORT	IRadarObjects
CameraControl	LaneDetection_Out	P-PORT	ILaneDetection
PerceptionManager	LaneDetection_In	R-PORT	ILaneDetection
PerceptionManager	CollisionRisk_Out	P-PORT	ICollisionRiskLegacy
BrakeAssistControl	CollisionRisk_In	R-PORT	ICollisionRisk
LaneKeepControl	SteeringCorrection_Out	P-PORT	ISteeringCorrection
PerceptionManager	SteeringCorrection_In	R-PORT	ISteeringCorrection
PerceptionManager	DriverWarning_Out	P-PORT	IDriverWarning
WarningManager	DriverWarning_In	R-PORT	IDriverWarning
CameraControl	CameraStatus_Out	P-PORT	ICameraStatus
WarningManager	CameraStatus_In	R-PORT	ICameraStatus

6. Signals

Signal	Data Type	Source	Destination
ObjectDistance	uint16	RadarControl	PerceptionManager
LaneOffset	int16	CameraControl	PerceptionManager
CollisionRisk	uint8	PerceptionManager	BrakeAssistControl
SteeringAngleCommand	int16	LaneKeepControl	RadarControl
WarningLevel	uint8	PerceptionManager	WarningManager
CameraStatus	boolean	CameraControl	WarningManager

7. Dependencies

Source	Relationship	Target	Reason
RadarControl	PROVIDES_INTERFACE	IRadarObjects	Publishes object distance.
PerceptionManager	REQUIRES_INTERFACE	IRadarObjects	Consumes object distance.
CameraControl	PROVIDES_INTERFACE	ILaneDetection	Publishes lane offset.
PerceptionManager	PROVIDES_INTERFACE	ICollisionRisk	Publishes collision risk.
BrakeAssistControl	REQUIRES_INTERFACE	ICollisionRisk	Consumes collision risk.
LaneKeepControl	PROVIDES_INTERFACE	ISteeringCorrection	Publishes steering angle command.
PerceptionManager	REQUIRES_INTERFACE	ISteeringCorrection	Consumes steering angle command.
PerceptionManager	PROVIDES_INTERFACE	IDriverWarning	Publishes warning level.
WarningManager	REQUIRES_INTERFACE	IDriverWarning	Consumes warning level.
CameraControl	PROVIDES_INTERFACE	ICameraStatus	Publishes camera status.
WarningManager	REQUIRES_INTERFACE	ICameraStatus	Consumes camera status.

8. Functional Flows

Flow F-001: Object Distance Distribution

RadarControl publishes ObjectDistance through IRadarObjects. PerceptionManager receives the signal through RadarObjects_In and updates its internal state model.

Flow F-002: Lane Offset Distribution

CameraControl publishes LaneOffset through ILaneDetection. PerceptionManager receives the signal through LaneDetection_In and updates its internal state model.

Flow F-003: Collision Risk Distribution

PerceptionManager publishes CollisionRisk through ICollisionRisk. BrakeAssistControl receives the signal through CollisionRisk_In and updates its internal state model.

Flow F-004: Steering Angle Command Distribution

LaneKeepControl publishes SteeringAngleCommand through ISteeringCorrection. PerceptionManager receives the signal through SteeringCorrection_In and updates its internal state model.

Flow F-005: Warning Level Distribution

PerceptionManager publishes WarningLevel through IDriverWarning. WarningManager receives the signal through DriverWarning_In and updates its internal state model.

Flow F-006: Camera Status Distribution

CameraControl publishes CameraStatus through ICameraStatus. WarningManager receives the signal through CameraStatus_In and updates its internal state model.

9. Integration Constraints

All provider and consumer ports shall reference compatible interfaces. Signal names shall remain consistent between interface definitions and functional flows. Architecture dependencies must be traceable to an approved source section.

10. Engineering Traceability

This synthetic HLD supports extraction, retrieval, validation and revision comparison experiments.